Overborrowing, Capital and Imperfect Information*

Chang Ma[a] Carlos Rondón-Moreno[b]

February 26, 2026

Abstract

Keywords: overborrowing; macroprudential policy; information frictions
JEL classification: D62, D84, E44, F32, F38, F41

*The views and conclusions presented in this paper are exclusively those of the authors and do not represent the views of the Central Bank of Chile or of its Board members.

[a] International School of Finance, Fudan University (FISF). *Email:* changma@fudan.edu.cn
[b] Central Bank of Chile. *E-mail:* crondon@bcentral.cl

1 Theoretical Framework

We build on the two-sector small open economy model with an occasionally binding collateral constraint in Benigno et al. (2013). Relative to their baseline framework, we enrich the shock structure by allowing for both permanent and transitory sectoral productivity shocks. We also relax the perfect-information assumption. Agents observe realized aggregate productivity but do not observe its underlying components, and must infer them using the information available in the economy. Productivity shocks are the only source of uncertainty.

1.1 Households

Households have standard constant relative risk aversion (CRRA) preferences over consumption and disutility from labor,

$$\mathbb{E}_0 \left[\sum_{t=0}^{\infty} \frac{\beta^t}{1-\sigma} \left(C_t - \frac{H_t^{\delta}}{\delta} \right)^{1-\sigma} \right], \qquad \sigma > 0, \tag{1}$$

where β is the subjective discount factor, σ denotes the inverse of the intertemporal elasticity of substitution, and δ governs the Frisch elasticity of labor supply.

Households can supply labor to both sectors. Let H_t^T and H_t^N denote labor in the tradable and non-tradable sectors, respectively, and let total labor be $H_t \equiv H_t^T + H_t^N$. We assume a common wage, W_t, across sectors. This assumption can be interpreted as the outcome of frictionless intratemporal labor mobility and a unified labor market, so that arbitrage equalizes wages across sectors in equilibrium.

Total consumption, C_t, is a CES bundle of tradable and non-tradable consumption, with elasticity of substitution ϵ,

$$C_t = \left[\omega \left(C_t^T \right)^{\frac{\epsilon-1}{\epsilon}} + (1-\omega) \left(C_t^N \right)^{\frac{\epsilon-1}{\epsilon}} \right]^{\frac{\epsilon}{\epsilon-1}},$$

where $\omega \in [0,1]$ governs the weight on tradable goods.

At the beginning of period t, households receive factor income and firm dividends,

repay outstanding debt, and choose consumption and next-period bond holdings. The period budget constraint, expressed in units of tradable goods, is

$$B_{t+1} = (1+r)B_t + W_t\left(H_t^T + H_t^N\right) + \Omega_t - C_t^T - p_t C_t^N, \tag{2}$$

where B_{t+1} denotes holdings of one-period, non-state-contingent foreign bonds to be repaid next period at the exogenous world interest rate r. Bonds are denominated in tradable goods. The relative price of non-tradables in terms of tradables is p_t, and the tradable good is the numeraire. The term Ω_t denotes aggregate firm profits (dividends) received by the household, measured in tradable units.

Foreign borrowing is limited by a collateral constraint that ties debt repayment capacity to contemporaneous resources in tradable units,

$$B_{t+1} \geq -\kappa\left[W_t\left(H_t^T + H_t^N\right) + \Omega_t\right], \tag{3}$$

where $\kappa > 0$ governs the tightness of the constraint.

The representative household chooses sequences $\{C_t^T, C_t^N, H_t^T, H_t^N, B_{t+1}\}_{t=0}^{\infty}$ to maximize lifetime utility in (1), subject to the aggregation of consumption into C_t given above, the budget constraint (2), the borrowing constraint (3), and the labor aggregation identity $H_t = H_t^T + H_t^N$. Taking prices $\{p_t\}$, wages $\{W_t\}$, firm dividends $\{\Omega_t\}$, and the world interest rate r as given, the household internalizes that labor supplied to either sector yields the same marginal remuneration because the wage is common across sectors.

Let Λ_t denote the multiplier on the budget constraint (2) and $\tilde{\mu}_t$ the multiplier on the borrowing constraint (3). The first-order conditions of this problem are as follows.

Optimal tradable consumption equates the marginal utility of tradable spending to the shadow value of wealth,

$$\omega\left(\frac{C_t}{C_t^T}\right)^{\frac{1}{\epsilon}}\left(C_t - \frac{H_t^\delta}{\delta}\right)^{-\sigma} = \Lambda_t. \tag{4}$$

Optimal non-tradable consumption equates the marginal utility of non-tradable spending,

converted to tradable units via p_t, to the shadow value of wealth,

$$(1-\omega)\left(\frac{C_t}{C_t^N}\right)^{\frac{1}{\epsilon}}\left(C_t-\frac{H_t^\delta}{\delta}\right)^{-\sigma}=\Lambda_t\,p_t. \tag{5}$$

Dividing (5) by (4) to eliminate Λ_t yields the relative price of non-tradables,

$$p_t=\frac{1-\omega}{\omega}\left(\frac{C_t^T}{C_t^N}\right)^{\frac{1}{\epsilon}}. \tag{6}$$

The labor supply condition, which is identical across sectors because the wage is common, balances the marginal disutility of work against its marginal value, inclusive of the collateral benefit when the borrowing constraint binds,

$$H_t^{\delta-1}\left(C_t-\frac{H_t^\delta}{\delta}\right)^{-\sigma}=(\Lambda_t+\kappa\,\tilde{\mu}_t)\,W_t. \tag{7}$$

Substituting Λ_t from (4) into (7) and dividing both sides by Λ_t gives the labor supply function,

$$H_t^{\delta-1}=\omega\left(\frac{C_t}{C_t^T}\right)^{\frac{1}{\epsilon}}\left(1+\kappa\,\frac{\tilde{\mu}_t}{\Lambda_t}\right)W_t. \tag{8}$$

The Euler equation for bond holdings is

$$\Lambda_t=\beta\,(1+r)\,\mathbb{E}_t\left[\Lambda_{t+1}\right]+\tilde{\mu}_t, \tag{9}$$

together with the complementary slackness conditions

$$\tilde{\mu}_t\geq 0,\qquad \tilde{\mu}_t\left(B_{t+1}+\kappa\left[W_t\left(H_t^T+H_t^N\right)+\Omega_t\right]\right)=0.$$

Labor supply and the anticipation of future constraints. When the borrowing constraint does not bind today ($\tilde{\mu}_t=0$) but is expected to bind tomorrow, the labor supply condition (8) is itself undistorted. However, the anticipation of a future binding constraint affects labor supply indirectly through the Euler equation. Iterating (9)

forward and using $\tilde{\mu}_t = 0$ gives

$$\Lambda_t = \beta(1+r)\left(\mathbb{E}_t[\tilde{\mu}_{t+1}] + \beta(1+r)\,\mathbb{E}_t[\Lambda_{t+2}]\right).$$

Higher expected future multipliers $\mathbb{E}_t[\tilde{\mu}_{t+s}]$ raise Λ_t, which through (4) lowers C_t^T and increases $\omega(C_t/C_t^T)^{1/\epsilon}$ in (8). The household therefore works more and consumes less today—building precautionary savings—even though the current-period labor wedge is zero.

1.2 Firms

There are two types of competitive firms: tradable-sector producers and non-tradable-sector producers. Each firm in sector $j \in \{T, N\}$ operates a decreasing-returns-to-scale technology using labor as the sole input,

$$Y_t^T = A_t^T \left(H_t^T\right)^{\alpha^T},$$
$$Y_t^N = A_t^N \left(H_t^N\right)^{\alpha^N},$$

where $\alpha^j \in (0,1)$ and A_t^j is sector-specific productivity.

Tradable-sector firms. The representative tradable firm takes the wage W_t as given and chooses labor to maximize profits measured in tradable units (the numeraire),

$$\max_{H_t^T} \; A_t^T \left(H_t^T\right)^{\alpha^T} - W_t H_t^T.$$

The first-order condition equates the marginal product of labor to the wage,

$$\alpha^T A_t^T \left(H_t^T\right)^{\alpha^T - 1} = W_t. \tag{10}$$

Tradable-sector profits are therefore

$$\Omega_t^T = Y_t^T - W_t H_t^T = (1-\alpha^T)\, Y_t^T.$$

Non-tradable-sector firms. The representative non-tradable firm takes the relative price p_t and the wage W_t as given and chooses labor to maximize profits in tradable units,

$$\max_{H_t^N} \; p_t \, A_t^N \left(H_t^N\right)^{\alpha^N} - W_t H_t^N.$$

The first-order condition is

$$p_t \, \alpha^N A_t^N \left(H_t^N\right)^{\alpha^N - 1} = W_t. \tag{11}$$

Non-tradable-sector profits, expressed in tradable units, are

$$\Omega_t^N = p_t \, Y_t^N - W_t H_t^N = (1 - \alpha^N) \, p_t \, Y_t^N.$$

Aggregate firm dividends received by the household are $\Omega_t = \Omega_t^T + \Omega_t^N$. Combining the two labor demand schedules (10) and (11) yields the relative price as a function of sectoral marginal products,

$$p_t = \frac{\alpha^T A_t^T \left(H_t^T\right)^{\alpha^T - 1}}{\alpha^N A_t^N \left(H_t^N\right)^{\alpha^N - 1}}.$$

Note that a decline in p_t reduces the tradable-unit revenue of each additional unit of non-tradable output, lowering the firm's labor demand. Because $Y_t^N = A_t^N (H_t^N)^{\alpha^N}$, non-tradable production falls. In general equilibrium the wage W_t is common across sectors, so a lower p_t reallocates labor away from the non-tradable sector toward the tradable sector (H_t^N falls, H_t^T rises), partially offsetting the contraction through the declining marginal cost of non-tradable labor.

1.3 Information Structure

To model the information problem, we build upon Boz et al. (2011) and Herreño and Rondón-Moreno (2025). Sectoral productivity decomposes into a common labor-augmenting

permanent component, Γ_t, and a sector-specific transitory component,

$$A_t^T = \Gamma_t \exp\left(z_t^T\right),$$
$$A_t^N = \Gamma_t \exp\left(z_t^N\right).$$

Let $\mathbf{z}_t \equiv \begin{bmatrix} z_t^T \\ z_t^N \end{bmatrix}$ follow a VAR(1),

$$\mathbf{z}_t = \mathbf{\Phi}\, \mathbf{z}_{t-1} + \boldsymbol{\varepsilon}_t,$$

with $\boldsymbol{\varepsilon}_t \sim \mathcal{N}(\mathbf{0},\, \mathbf{\Sigma}_\varepsilon)$. The VAR(1) is stable when all eigenvalues of $\mathbf{\Phi}$ lie strictly inside the unit circle. Equivalently, $\rho(\mathbf{\Phi}) < 1$, where $\rho(\mathbf{\Phi}) \equiv \max\{|\lambda| \mid \lambda \text{ is an eigenvalue of } \mathbf{\Phi}\}$.

The permanent component evolves as

$$\Gamma_t = \Gamma_{t-1} \exp(g_t) = \prod_{j=0}^{t} \exp(g_j),$$

where g_t is the stochastic growth rate. We assume

$$g_t = (1-\rho_g)\mu_g + \rho_g g_{t-1} + \varepsilon_t^g,$$

with $|\rho_g| < 1$ and $\varepsilon_t^g \sim \mathcal{N}(0, \sigma_g^2)$.

Learning Problem Agents do not observe the decomposition of productivity into Γ_t and $\mathbf{z}_t$. Instead, they observe realized sectoral productivity and infer its latent components using the information available at time t. We assume that households know the parameters governing the productivity processes and observe the full history of realized productivity in both sectors. In addition to realized productivity, agents receive a noisy signal about the permanent growth rate,

$$\tilde{s}_t = g_t + \nu_t, \qquad \nu_t \sim \mathcal{N}(0, \sigma_\nu^2),$$

where ν_t is i.i.d. and independent of all other shocks. The precision of this signal, governed by σ_ν^2, parameterizes the quality of information available to agents. When $\sigma_\nu^2 \to 0$ the signal is perfectly informative and the permanent component is observed; as $\sigma_\nu^2 \to \infty$ the signal becomes uninformative and agents must rely entirely on realized productivity to infer g_t.[1]

Given the linear-Gaussian structure, households form beliefs using the Kalman filter, which solves an optimal signal-extraction problem in which the latent state is updated by combining a model-based forecast with the information contained in newly observed signals.

Let $\mathbb{I}_t$ denote the household information set,

$$\mathbb{I}_t \equiv \left\{ \left\{ A_{t-s}^i,\, \tilde{s}_{t-s} \right\}_{s=0}^{\infty},\, f\left(\varepsilon_t^T,\, \varepsilon_t^N\right),\, f(\varepsilon_t^g),\, f(\nu_t) \right\}, \qquad \forall i \in \{T, N\},$$

where $\{A_{t-s}^i\}_{s=0}^{\infty}$ is the history of realized productivity, $\{\tilde{s}_{t-s}\}_{s=0}^{\infty}$ is the history of noisy signals, and $f(\cdot)$ denotes the innovation distributions.

To connect observables to the latent states, define the growth rate of tradable productivity and the relative growth rate of non-tradable productivity,

$$\Psi_t^T \equiv \ln\left(\frac{A_t^T}{A_{t-1}^T}\right) = \left(z_t^T - z_{t-1}^T\right) + g_t, \tag{12}$$

$$\Psi_t^N \equiv \ln\left(\frac{A_t^N}{A_{t-1}^T}\right) = \left(z_t^N - z_{t-1}^T\right) + g_t. \tag{13}$$

[1] When $\sigma_\nu^2 = 0$ the signal reveals g_t exactly. Given g_t, the productivity-growth observables identify $z_t^T - z_{t-1}^T = \Delta_t^T - g_t$ and $z_t^N - z_{t-1}^T = \Delta_t^N - g_t$, so all latent components are recovered and the model reduces to the perfect-information benchmark.

Let $s_t \equiv \begin{bmatrix} \Psi_t^T \\ \Psi_t^N \\ \tilde{s}_t \end{bmatrix}$ collect the observable signals and define the state vector

$$\alpha_t \equiv \begin{bmatrix} z_t^T \\ z_t^N \\ g_t \\ z_{t-1}^T \end{bmatrix}.$$

Then the observation equation is

$$s_t = \mathbf{Z}\alpha_t + \boldsymbol{\xi}_t, \qquad \mathbf{Z} \equiv \begin{bmatrix} 1 & 0 & 1 & -1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & 0 \end{bmatrix}, \tag{14}$$

where $\boldsymbol{\xi}_t \equiv \begin{bmatrix} 0 & 0 & \nu_t \end{bmatrix}'$ is the observation noise vector with covariance

$$\mathbf{H} \equiv \mathrm{Var}(\boldsymbol{\xi}_t) = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \sigma_\nu^2 \end{bmatrix}.$$

The first two rows of s_t are observed without error, while the third row is contaminated by the signal noise ν_t.

The transition equation is

$$\alpha_t = \mathbf{c} + \mathbf{\Pi}\alpha_{t-1} + \mathbf{R}\eta_t, \tag{15}$$

where

$$\mathbf{c} \equiv \begin{bmatrix} 0 \\ 0 \\ (1-\rho_g)\mu_g \\ 0 \end{bmatrix}, \qquad \mathbf{\Pi} \equiv \begin{bmatrix} \rho_{z^T,z^T} & \rho_{z^T,z^N} & 0 & 0 \\ \rho_{z^N,z^T} & \rho_{z^N,z^N} & 0 & 0 \\ 0 & 0 & \rho_g & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}, \qquad \mathbf{R} \equiv \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix},$$

and $\eta_t \equiv \begin{bmatrix} \varepsilon_t^T & \varepsilon_t^N & \varepsilon_t^g \end{bmatrix}'$ satisfies

$$\eta_t \sim \mathcal{N}(\mathbf{0},\, \mathbf{Q}), \qquad \mathbf{Q} \equiv \begin{bmatrix} \sigma^2_{z^T} & \sigma_{z^T,z^N} & 0 \\ \sigma_{z^N,z^T} & \sigma^2_{z^N} & 0 \\ 0 & 0 & \sigma^2_g \end{bmatrix}.$$

Let $a_{t|t-1} \equiv \mathbb{E}[\alpha_t \mid \mathbb{I}_{t-1}]$ and $a_t \equiv \mathbb{E}[\alpha_t \mid \mathbb{I}_t]$ denote the prior and posterior state estimates. The Kalman filter updates beliefs in two steps. The prediction step uses the transition equation to map last period's posterior into a prior for the current state and an implied forecast for the signals. The updating step then compares the observed signals to this forecast and adjusts beliefs proportionally to the resulting innovation, placing more weight on the data when the innovation is relatively precise and more weight on the model-based forecast when the signals are less informative. Formally,

$$a_{t|t-1} = \mathbf{c} + \mathbf{\Pi} a_{t-1}, \tag{16}$$

$$a_t = a_{t|t-1} + \mathbf{K}\left(s_t - \mathbf{Z} a_{t|t-1}\right), \tag{17}$$

where the Kalman gain is $\mathbf{K} \equiv \mathbf{P}\mathbf{Z}'(\mathbf{Z}\mathbf{P}\mathbf{Z}' + \mathbf{H})^{-1}$ and $\mathbf{P}$ is the steady-state forecast-error covariance matrix solving the discrete-time algebraic Riccati equation,

$$\mathbf{P} = \mathbf{\Pi}\mathbf{P}\mathbf{\Pi}' - \mathbf{\Pi}\mathbf{P}\mathbf{Z}'(\mathbf{Z}\mathbf{P}\mathbf{Z}' + \mathbf{H})^{-1}\mathbf{Z}\mathbf{P}\mathbf{\Pi}' + \mathbf{R}\mathbf{Q}\mathbf{R}'. \tag{18}$$

Equations (16)–(17) summarize belief updating: the posterior combines the prior with the innovation $s_t - \mathbf{Z} a_{t|t-1}$, with weights determined endogenously by $\mathbf{K}$. The observation-

noise matrix $\mathbf{H}$ enters the Kalman gain and the Riccati equation, so that more precise signals (σ_ν^2 small) place greater weight on the noisy signal relative to the model-based forecast, while imprecise signals (σ_ν^2 large) lead agents to rely primarily on realized productivity growth.

1.4 Competitive Equilibrium

Because total factor productivity contains the permanent component Γ_t, quantities that are proportional to technology—consumption, output, wages, bonds, and profits—inherit a stochastic trend and are therefore non-stationary. To render the model stationary we normalize all trending variables by the lagged level of tradable productivity, $A_{t-1}^T = \Gamma_{t-1}\exp(z_{t-1}^T)$. For any trending variable X_t, define its detrended counterpart in lowercase as

$$x_t \equiv \frac{X_t}{A_{t-1}^T}.$$

The Lagrange multiplier Λ_t on the budget constraint is the marginal utility of tradable wealth, which scales with $(A_{t-1}^T)^{-\sigma}$ along the balanced growth path. We therefore define the stationary multipliers

$$\lambda_t \equiv \Lambda_t \left(A_{t-1}^T\right)^{-\sigma}, \qquad \mu_t \equiv \tilde{\mu}_t \left(A_{t-1}^T\right)^{-\sigma}. \tag{19}$$

Detrended equilibrium system. Using these definitions, denoting the gross growth rate of tradable productivity by $\gamma_t^T \equiv A_t^T/A_{t-1}^T = \exp(\Psi_t^T)$, and noting that $A_t^j/A_{t-1}^T = \exp(\Psi_t^j)$ from the signal definitions (12)–(13), we can write the detrended production functions as $y_t^T = \exp(\Psi_t^T)(H_t^T)^{\alpha^T}$ and $y_t^N = \exp(\Psi_t^N)(H_t^N)^{\alpha^N}$. Three market clearing conditions close the model: non-tradable goods market clearing ($c_t^N = y_t^N$), labor market clearing ($H_t = H_t^T + H_t^N$), and aggregate output ($y_t = y_t^T + p_t y_t^N$). After imposing market clearing and substituting out profits—using $\pi_t = (1-\alpha^T)y_t^T + (1-\alpha^N)p_t y_t^N$—and wages from the firm first-order conditions, the detrended competitive equilibrium reduces to the following system.

The shadow value of tradable wealth is

$$\omega \left(\frac{c_t}{c_t^T}\right)^{\frac{1}{\epsilon}} \left(c_t - \frac{H_t^\delta}{\delta}\right)^{-\sigma} = \lambda_t. \tag{20}$$

The relative price of non-tradables is

$$p_t = \frac{1-\omega}{\omega} \left(\frac{c_t^T}{c_t^N}\right)^{\frac{1}{\epsilon}}. \tag{21}$$

Non-tradable market clearing pins down non-tradable consumption,

$$c_t^N = \exp(\Psi_t^N)(H_t^N)^{\alpha^N}. \tag{22}$$

The wage is determined by the tradable firm's first-order condition,

$$w_t = \alpha^T \exp(\Psi_t^T)(H_t^T)^{\alpha^T - 1}. \tag{23}$$

Equating labor demands across sectors yields

$$\alpha^T \exp(\Psi_t^T)(H_t^T)^{\alpha^T-1} = p_t\, \alpha^N \exp(\Psi_t^N)(H_t^N)^{\alpha^N-1}. \tag{24}$$

The labor supply function is

$$H_t^{\delta-1} = \omega \left(\frac{c_t}{c_t^T}\right)^{\frac{1}{\epsilon}} \left(1 + \kappa\, \frac{\mu_t}{\lambda_t}\right) w_t, \tag{25}$$

where total labor satisfies $H_t = H_t^T + H_t^N$. The Euler equation is

$$\lambda_t = \beta(1+r)\, \mathbb{E}_t\Big[\left(\gamma_t^T\right)^{-\sigma} \lambda_{t+1}\Big] + \mu_t, \tag{26}$$

where the factor $(\gamma_t^T)^{-\sigma}$ captures the change in the normalization base between periods t and $t+1$. The budget constraint, after substituting wage payments and profits, becomes

$$b_{t+1}\, \gamma_t^T = (1+r)b_t + y_t^T - c_t^T \tag{27}$$

The borrowing constraint is

$$b_{t+1}\,\gamma_t^T \geq -\kappa\left[y_t^T + p_t\,y_t^N\right], \tag{28}$$

together with $\mu_t \geq 0$ and the complementary slackness condition.

A *competitive equilibrium* is a set of detrended allocations $\{c_t^T, c_t^N, H_t^T, H_t^N, b_{t+1}\}_{t=0}^{\infty}$ and prices $\{p_t, w_t\}_{t=0}^{\infty}$ such that conditions (20)–(28) hold and beliefs are updated according to the Kalman filter (16)–(17), given the information set $\mathbb{I}_t$.

1.5 Social Planner

The competitive equilibrium is constrained inefficient because individual households take the relative price p_t as given when making borrowing decisions. They therefore fail to internalize the following pecuniary externality: a collective increase in borrowing today reduces tradable consumption when the constraint binds, which—through the pricing rule (21)—depresses the relative price of non-tradables, tightening the constraint precisely when agents need to borrow most. A constrained social planner corrects this externality while remaining subject to the same technological and informational constraints as private agents.

Recursive formulation. The planner's problem is recursive. Let the state vector be $\mathbf{x}_t \equiv (b_t,\, a_t)$, where b_t are current detrended bond holdings and $a_t = \mathbb{E}[\alpha_t \mid \mathbb{I}_t]$ is the Kalman posterior at time t. Conditional on $\mathbf{x}_t$, the planner observes the current-period signals (Ψ_t^T, Ψ_t^N), which determine sectoral productivity, and takes as constraints the competitive equilibrium pricing rule, the non-tradable market clearing $c_t^N = y_t^N(H_t^N)$, and the labor market clearing condition. Because non-tradable consumption is pinned down by H_t^N through $c_t^N = \exp(\Psi_t^N)(H_t^N)^{\alpha^N}$, the planner chooses $\{c_t^T, H_t^T, H_t^N, b_{t+1}\}$ to solve

$$V(\mathbf{x}_t) = \max_{c_t^T,\, H_t^T,\, H_t^N,\, b_{t+1}} \left\{ \frac{1}{1-\sigma}\left(c_t - \frac{H_t^{\delta}}{\delta}\right)^{1-\sigma} + \beta\,\mathbb{E}_t[V(\mathbf{x}_{t+1})] \right\}, \tag{29}$$

where $c_t = [\omega(c_t^T)^{(\epsilon-1)/\epsilon} + (1-\omega)(c_t^N)^{(\epsilon-1)/\epsilon}]^{\epsilon/(\epsilon-1)}$ with $c_t^N = \exp(\Psi_t^N)(H_t^N)^{\alpha^N}$, and $\mathbf{x}_{t+1} = (b_{t+1},\, a_{t+1})$ evolves via the Kalman filter (16)–(17). The problem is subject to the tradable resource constraint (27) (reproduced here for clarity),

$$b_{t+1}\,\gamma_t^T = (1+r)b_t + \exp(\Psi_t^T)(H_t^T)^{\alpha^T} - c_t^T, \tag{30}$$

and the aggregate borrowing constraint,

$$b_{t+1}\,\gamma_t^T \;\geq\; -\kappa\left[\exp(\Psi_t^T)(H_t^T)^{\alpha^T} + \frac{1-\omega}{\omega}\,(c_t^T)^{\frac{1}{\epsilon}}\left(\exp(\Psi_t^N)(H_t^N)^{\alpha^N}\right)^{\frac{\epsilon-1}{\epsilon}}\right], \tag{31}$$

Notice the collateral constraint is expressed entirely in terms of the planner's choice variables (c_t^T, H_t^T, H_t^N) reflecting the direct effect of H_t^N on non-tradable output and the indirect effect of c_t^T on the relative price.

First-order conditions. Let λ_t^{SP} and μ_t^{SP} denote the multipliers on (30) and (31), respectively. The first order conditions associated to the sequential version of (29)–(31) are

$$\omega\left(\frac{c_t}{c_t^T}\right)^{\frac{1}{\epsilon}}\left(c_t - \frac{H_t^\delta}{\delta}\right)^{-\sigma} = \lambda_t^{SP} + \mu_t^{SP}\,\Xi_t, \tag{32}$$

$$H_t^{\delta-1}\left(c_t - \frac{H_t^\delta}{\delta}\right)^{-\sigma} = \left(\lambda_t^{SP} + \kappa\mu_t^{SP}\right)\alpha^T\exp(\Psi_t^T)(H_t^T)^{\alpha^T-1}, \tag{33}$$

$$\begin{aligned} H_t^{\delta-1}\left(c_t - \frac{H_t^\delta}{\delta}\right)^{-\sigma} = \Bigg[&(1-\omega)\left(\frac{c_t}{\exp(\Psi_t^N)(H_t^N)^{\alpha^N}}\right)^{\frac{1}{\epsilon}}\left(c_t - \frac{H_t^\delta}{\delta}\right)^{-\sigma} \\ &+ \kappa\mu_t^{SP}\,p_t\,\frac{\epsilon-1}{\epsilon}\Bigg]\,\alpha^N\exp(\Psi_t^N)(H_t^N)^{\alpha^N-1}, \end{aligned} \tag{34}$$

$$\lambda_t^{SP} = \beta(1+r)\,\mathbb{E}_t\left[\left(\gamma_t^T\right)^{-\sigma}\lambda_{t+1}^{SP}\right] + \mu_t^{SP}, \tag{35}$$

$$\mu_t^{SP} \geq 0,$$

$$\begin{aligned} \mu_t^{SP}\Big(b_{t+1}\gamma_t^T + \kappa\Big[&\exp(\Psi_t^T)(H_t^T)^{\alpha^T} \\ &+ \frac{1-\omega}{\omega}(c_t^T)^{\frac{1}{\epsilon}}\left(\exp(\Psi_t^N)(H_t^N)^{\alpha^N}\right)^{\frac{\epsilon-1}{\epsilon}}\Big]\Big) = 0. \end{aligned}$$

where Ξ_t denotes the elasticity of the collateral constraint with respect to tradable consumption and is given by

$$\Xi_t \equiv \frac{\partial CC_t}{\partial c_t^T} = -\frac{\kappa}{\epsilon} \cdot \frac{p_t \exp(\Psi_t^N)(H_t^N)^{\alpha^N}}{c_t^T}, \tag{36}$$

where the equality uses $\partial p_t / \partial c_t^T = p_t / (\epsilon c_t^T)$ from (21).

References

Benigno, G., H. Chen, C. Otrok, A. Rebucci, and E. R. Young (2013). Financial crises and macroprudential policies. Journal of International Economics 89(2), 453 – 470.

Boz, E., C. Daude, and C. B. Durdu (2011). Emerging market business cycles: Learning about the trend. Journal of Monetary Economics 58(6), 616–631.

Herreño, J. and C. Rondón-Moreno (2025). Overborrowing and systemic externalities in the business cycle under imperfect information. Journal of International Economics 157, 104103.